Week 3

1 From Extracting Features to Evaluating Them

Week 1 introduced pretrained DINOv2 embeddings for DESI galaxy images. Week 2 explored a small DINO-style student-teacher training pipeline. Week 3 returns to pretrained models and asks whether their representations are useful for astronomical images. The work has two parts: inspecting attention as a simple segmentation signal, and measuring galaxy morphology classification using frozen features.

The two notebooks implement these experiments. This report explains their methods and evaluation protocol. No completed Week 3 execution outputs were available when it was prepared, so it does not state measured accuracy, a winning layer, or observed segmentation performance.

2 What Changes in DINOv2

DINOv2 extends the DINO approach with image-level and patch-level learning. Its masked patch objective, derived from iBOT, encourages the student to predict teacher representations for masked patches. This is relevant because segmentation needs local information rather than only one representation of the entire image.

The training recipe also includes KoLeo regularization, which encourages feature diversity, and Sinkhorn-Knopp normalization of teacher targets. Large-scale curated data and model distillation help produce strong pretrained representations. Week 3 uses these pretrained weights; it does not reproduce the DINOv2 training process.

3 Load the Galaxy Images

The attention notebook uses RGB cutouts from the MultimodalUniverse GZ10 dataset by default. It also accepts raw Week 1 g/r/z arrays or an image folder. For raw g/r/z inputs, the display conversion maps z, r and g to red, green and blue using a Lupton stretch. The original Week 1 conversion should be reused if an exact preprocessing match is required.

A contact sheet supports selecting around ten representative cutouts. The selection should include isolated galaxies, blended sources, foreground stars and extended or irregular objects. These categories must be checked visually rather than assumed from the first ten entries.

4 Image Preprocessing

Each RGB image is resized to 224 by 224 pixels, converted into a tensor and normalized with ImageNet mean and standard deviation. A batch dimension is added before inference. The backbone is placed in evaluation mode and gradients are disabled, so attention extraction does not update its parameters.

Input tensor for one image: (1, 3, 224, 224).

5 DINO ViT-S/8 Attention Maps

The first comparison uses pretrained DINO ViT-S/8. An 8-pixel patch size gives a 28 by 28 grid at 224-pixel resolution, producing 784 patch tokens plus one CLS token. The last-layer attention tensor for one image has shape (1, 6, 785, 785), corresponding to batch, heads, query tokens and key tokens.

The notebook selects the CLS query row and removes the CLS key column. The remaining 784 values in each head are reshaped into a 28 by 28 map and enlarged to the image size using nearest-neighbour interpolation. Each head is displayed separately because heads may emphasize different spatial structures.

6 Build Naive Masks from Attention

For each head, patch attention values are normalized over the patch tokens and sorted from largest to smallest. The mask retains the smallest leading set whose cumulative mass reaches at least 60%. The patch that crosses the threshold is included. This is a threshold on cumulative attention mass, not a choice to retain 60% of all patches.

The individual head masks are saved alongside the attention maps. Their union is overlaid on the RGB image for a compact display. A union can include many sources, so it should not be treated as a validated galaxy segmentation method.

Retain the smallest k such that the sum of the k largest normalized scores is at least 0.60.

7 Compare DINOv2 With and Without Registers

The notebook repeats the analysis using DINOv2 ViT-S/14 and its register-token variant. A 14-pixel patch size produces a 16 by 16 grid with 256 patch tokens. This coarser grid should be considered when comparing masks with the finer DINO ViT-S/8 grid.

DINOv2 does not provide the same get_last_selfattention method. A hook captures the input to its final attention module and reconstructs the CLS attention probabilities from the query and key projections. All key tokens participate in the softmax; CLS and register columns are removed before the patch-only mask is computed.

8 Registers and Segmentation Limitations

Registers are additional learnable tokens intended to absorb internal global-information storage that can otherwise appear in background patches. The comparison checks whether the register checkpoint produces fewer isolated attention spikes. This is a question to evaluate from the generated maps, not a result already established by the notebook.

Attention may miss faint outskirts, join blended galaxies or highlight stars and neighbouring objects. The notebook records mask area and peak patch attention as diagnostics. Without verified ground-truth masks, these values do not establish segmentation accuracy, IoU or Dice score.

9 Galaxy Zoo Morphology Labels and the Data Split

The probe notebook loads GZ10 images and morphology labels, using the rgb_image and gz10_label fields. The assignment describes approximately 17,736 galaxies in ten classes. The notebook prints the actual loaded size and class distribution before evaluation, because imbalanced class counts can make overall accuracy misleading.

A seeded, stratified 80/20 split creates development and test sets. The development set is split again for training and validation. Feature and probe selection use validation accuracy, with macro F1 as a tie-breaker. Final models are refitted on the full development set and evaluated on the same held-out test set.

10 Extract and Cache the Last Four Layers

The frozen DINOv2 ViT-S/14 backbone extracts intermediate representations from layers 9 to 12. For each image and layer, it returns 256 patch vectors of dimension 384 and one CLS vector of dimension 384. On a GPU, feature extraction uses automatic mixed precision to reduce computation and memory requirements.

CLS vectors, final-layer mean-patch vectors and, by default, all four layers of patch tokens are saved as NumPy files. Full patch caching requires about 14 GB for the complete catalog in float16. A cache manifest and committed-row counter support reuse and interrupted-run recovery. Disabling patch storage makes the run lighter but does not satisfy the full-token caching requirement.

11 Build the Six Feature Sets

Four feature sets use the individual CLS vectors from layers 9, 10, 11 and 12. A fifth averages the 256 patch vectors from layer 12 into one 384-dimensional vector. The sixth concatenates the four CLS vectors into a 1,536-dimensional vector.

This comparison tests whether an earlier layer, local-token averaging or a combination of depths represents morphology better than the final CLS vector. The winning representation must be identified from validation results rather than assumed in advance.

Individual CLS: 384 dimensions | Mean patch: 384 | Concatenated CLS: 1,536

12 Train Linear and MLP Probes

Two classifiers are trained on every feature set. Logistic regression is the linear probe: it measures how well morphology classes can be separated directly in the frozen feature space. The MLP uses one hidden layer with 256 units, ReLU activation and a ten-class output layer. It is trained with Adam for 30 epochs using batches of 64.

Features are standardized before training. The scaler is fitted only on the relevant training partition and then applied to validation or test features. The DINOv2 backbone remains frozen throughout. Both probes use the same split, and the baseline does not apply class weighting.

13 Evaluation, Confusion Matrices and t-SNE

The notebooks produce a feature-set-by-probe results table, overall accuracy, balanced accuracy, macro F1 and per-class metrics. Per-class accuracy is the fraction of images from a true class that are predicted correctly, equivalent to that class's recall. The confusion matrix for the validation-selected setting shows counts and row-normalized errors.

The t-SNE comparison uses the final-layer CLS features and the best individual CLS layer chosen on validation. Both views use the same stratified test-image subset. Standardization and PCA are fitted on development images before visualization. Cluster separation is exploratory and does not replace classification metrics.

14 Comparison With the Assignment Reference

The Week 3 brief cites morphology accuracies of about 67% for DINOv2 ViT-S/14 and 73% for ViT-L/14 with an MLP on CLS embeddings. These are reference values from the brief, not results obtained in this work. The implemented quantitative experiment uses ViT-S/14; it does not evaluate ViT-L/14.

The closest implemented comparison is the layer-12 CLS feature set with the MLP. Any measured difference must be interpreted alongside the dataset split, RGB preprocessing, checkpoint and classifier recipe. A single seeded run does not measure variability across training seeds.

15 Questions the Experiments Must Answer

Main galaxy or every bright source? Compare the maps with the RGB cutouts, checking stars and neighbouring galaxies. **Are the masks usable?** Inspect faint outskirts, blending and fragmentation; visual usefulness alone does not establish quantitative segmentation quality.

Which representation is best? Read the validation table and report the selected setting's held-out test performance. **Which classes are confused, and why?** Use the confusion matrix and inspect examples before assigning a visual explanation. **How well do natural-image features transfer?** Assess held-out morphology accuracy and per-class performance together with the qualitative attention analysis. The available unexecuted notebooks do not yet settle these questions.

16 Results and Limitations

The deliverables currently provide the complete attention and probing procedures. Verified Week 3 maps, class counts, accuracy tables and embedding plots were not available for this report. Consequently, no numerical outcome or observed advantage of registers is claimed. These findings should be added after executing the notebooks and inspecting their saved outputs.

The Big Picture

Galaxy cutout to pretrained DINO/DINOv2 to attention maps and cached embeddings to masks, morphology probes and visualizations. Week 3 moves from extracting representations to testing their spatial behaviour and their usefulness for galaxy morphology.

Basis: Week3.pdf; Week3_Attention_Maps.ipynb; Week3_Layerwise_Probe.ipynb.
Structure and explanatory style follow the supplied Week 1 and Week 2 reports.